

Exercise Solution

4.1 Write the simplest “AND-OR” functions of the following logic functions by K-map.

4.2 Detect and eliminate the hazard of the following logic functions.

$$(a) F = WX + \overline{W} \cdot \overline{Y} + \overline{X}Y \quad (b) F = W\overline{X} \cdot \overline{Y} + X\overline{Y}Z$$

(a) Let “W” be “1”, and “Y” be “1”. We have:

$$F = WX + \overline{W} \cdot \overline{Y} + \overline{X}Y = 1 \cdot X + \overline{1} \cdot \overline{1} + \overline{X} \cdot 1 = X + \overline{X}$$

The function “F” has the hazard caused by “X” and “ $\overline{X}$ ”. We add the redundant term “WY” to eliminate the hazard.

Let “W” be “0”, and “X” be “0”. We have:

$$F = WX + \overline{W} \cdot \overline{Y} + \overline{X}Y = 0 \cdot 0 + \overline{0} \cdot \overline{Y} + \overline{0} \cdot Y = \overline{Y} + Y$$

The function “F” has the hazard caused by “Y” and “ $\overline{Y}$ ”. We add the redundant term “ $\overline{W} \cdot \overline{X}$ ” to eliminate the hazard.

Let “X” be “1”, and “Y” be “0”. We have:

$$F = WX + \overline{W} \cdot \overline{Y} + \overline{X}Y = W \cdot 1 + \overline{W} \cdot \overline{0} + \overline{1} \cdot 0 = W + \overline{W}$$

The function “F” has the hazard caused by “W” and “ $\overline{W}$ ”. We add the redundant term “ $X \cdot \overline{Y}$ ” to eliminate the hazard.

Therefore, we add the three redundant terms “WY”, “ $\overline{W} \cdot \overline{X}$ ” and “ $X \cdot \overline{Y}$ ” to the original logic function.

$$F = WX + \overline{W} \cdot \overline{Y} + \overline{X}Y = WX + \overline{W} \cdot \overline{Y} + \overline{X}Y + WY + \overline{W} \cdot \overline{X} + X \cdot \overline{Y}$$

(b) Let “W” be “1”, “Y” be “0”, and “Z” be “1”. We have:

$$\begin{aligned} F &= W\overline{X} \cdot \overline{Y} + X\overline{Y}Z \\ &= \overline{X} \cdot (W\overline{Y}) + X \cdot (\overline{Y}Z) \\ &= \overline{X} \cdot (1 \cdot \overline{0}) + X \cdot (\overline{0} \cdot 1) \\ &= \overline{X} + X \end{aligned}$$

The function “F” has the hazard caused by “X” and “ $\overline{X}$ ”. We add the redundant term “ $(WY) \cdot (\overline{Y}Z) = WYZ$ ” to eliminate the hazard.

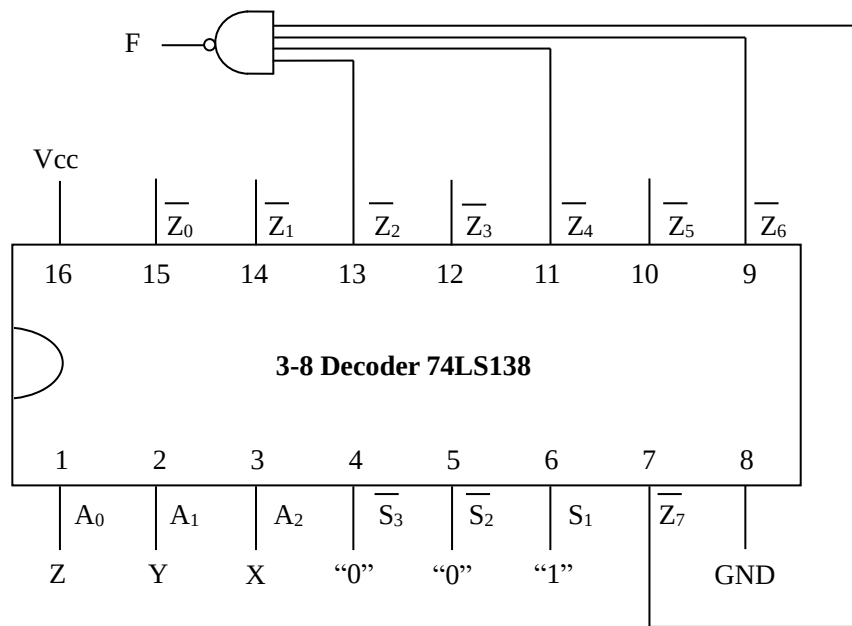
Therefore, we add the redundant term “ WYZ ” to the original logic function.

$$\begin{aligned}
 F &= W\overline{X} \bullet \overline{Y} + X\overline{Y}Z \\
 &= W\overline{X} \bullet \overline{Y} + X\overline{Y}Z + W\overline{Y}Z
 \end{aligned}$$

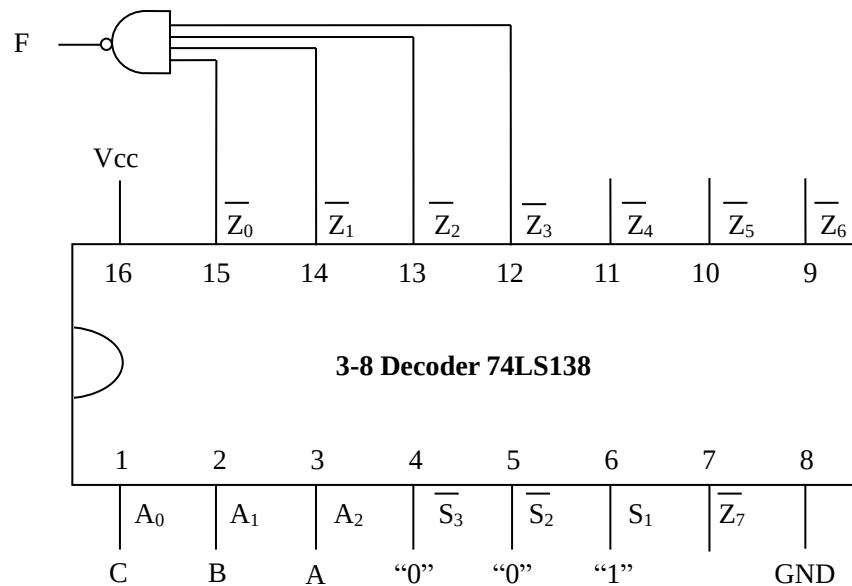
4.3 Use the 3-8 decoder and the “NAND” gate to implement the following logic functions.

(a) $F(X, Y, Z) = \sum m(2, 4, 6, 7)$ (b) $F(A, B, C) = \prod M(4, 5, 6, 7)$

(a) $F(X, Y, Z) = \sum m(2, 4, 6, 7) = \overline{\overline{m_2} \bullet \overline{m_4} \bullet \overline{m_6} \bullet \overline{m_7}}$



(b) $F(A, B, C) = \prod M(4, 5, 6, 7) = \sum m(0, 1, 2, 3) = \overline{\overline{m_0} \bullet \overline{m_1} \bullet \overline{m_2} \bullet \overline{m_3}}$



4.4 Use the 4-to-1 line data selector to implement the following logic functions.

(a) $F(A, B, C, D) = \sum m(0, 2, 4, 10, 12)$

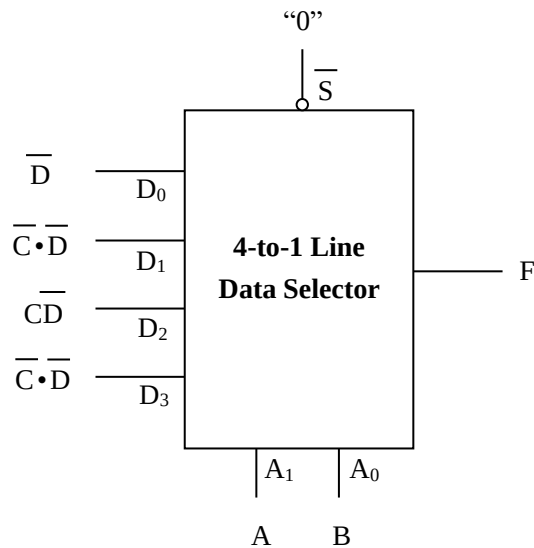
(b) $F(W, X, Y, Z) = \sum m(2, 3, 4, 5, 10, 12, 14)$

(a) $F(A, B, C, D) = \sum m(0, 2, 4, 10, 12)$

		CD			
		00	01	11	10
F	AB				
	00	1	0	0	1
	01	1	0	0	0
	11	1	0	0	0
	10	0	0	0	1

$$F = \bar{A} \cdot \bar{B} \cdot \bar{D} + \bar{A} B \cdot \bar{C} \cdot \bar{D} + A \bar{B} \cdot C \bar{D} + A B \cdot \bar{C} \cdot \bar{D}$$

$$D_0 = \bar{D} \quad D_1 = \bar{C} \cdot \bar{D} \quad D_2 = C \bar{D} \quad D_3 = \bar{C} \cdot \bar{D}$$



(b) $F(W, X, Y, Z) = \sum m(2, 3, 4, 5, 10, 12, 14)$

F \ YZ	WX			
	00	01	11	10
00	0	0	1	1
01	1	1	0	0
11	1	0	0	1
10	0	0	0	1

$$F = \overline{W} \cdot \overline{X} \cdot Y + \overline{W}X \cdot \overline{Y} + W\overline{X} \cdot Y\overline{Z} + WX \cdot \overline{Z}$$

$$D_0 = Y \quad D_1 = \overline{Y} \quad D_2 = Y\overline{Z} \quad D_3 = \overline{Z}$$

